# The cyclic-49 kernel, in the base field

CT33 independent Python + Go development replay | F_101 | November 12, 2026 UTC

## The construction

Two nonbacktracking 7-isogenies E0 -> E1 -> E2 compose to a separable degree-49 map. We compute the 24 x-coordinate pairs of its nonzero cyclic-49 kernel using only F_101 polynomial arithmetic; no F_(101^21) arithmetic is required to write the kernel polynomial.

## Prime-field formulas

Write X=N/D with D=h0^2. From the paired Velu sums, N=(7x-2S1)D - 2f'(x)h0'(x)h0(x) + 4f(x)(h0'(x)^2-h0(x)h0''(x)), where S1 is the sum of roots of h0.

If h1(X)=X^3+c2 X^2+c1 X+c0, set H=N^3+c2 N^2 D+c1 N D^2+c0 D^3. Then K=h0 H has degree 24 and the reduced composite x-map denominator is K^2.

## Constructive orbit results

### 109-point component

H irreducible degree 21; x-support field F_(101^21); E2=(59,27).

### 113-point component

H factors 7 + 7 + 7; x-support field F_(101^7); E2=(23,7).

These are support-polynomial facts. Full signed kernel points still require degree 42 and 14 extensions respectively; prime-field coefficients do not make individual points rational.

AUDIT / LIMITATIONS

# What was actually checked

Raw scripts, coefficient lists, commands and failed-development control in CT33_XONLY_KERNEL.md

## Independent local execution

- Python stdlib and separately implemented Go both reconstruct both first and second 7-isogeny x-maps.
- Both pass global elliptic rational-function identities; deliberately wrong codomain b-coefficients fail.
- The composed degree-49 rational function has a reduced numerator degree 49 and denominator K(x)^2.
- Both implementations compute all Frobenius gcds for primitive degree-21 supports.
- Selected nondual H is squarefree and disjoint from h0; the previous-dual/backtracking support differs.
- An independent SymPy factor_list spot check returns irreducible 21 and three irreducible 7 factors.
- Python -O and ordinary mode produce byte-for-byte identical output; checks do not depend on assert.

## Proof obligation / negative result

Cyclicity follows because the second kernel is not the first map's dual kernel. The extension-field orbit predictions alone would not establish that fact. Backtracking has 3/6-degree Frobenius orbits; demanding that its degree-21 pullback split over F_(101^21) was an incorrect early test and has been removed.

## Remaining acceptance work

Reconcile K with the actual two-hop Jacobian correspondence, check full independent Cantor addition and cross-map membership, obtain signed PCC, and run admitted cold complete solving-cost comparisons including optimized rho and existing index calculus. No paid compute launched; no speedup claimed.